

Cybersecurity Dashboard - Project Proposal

Project Overview : A Cybersecurity Dashboard that helps security teams monitor, detect, and analyze security incidents in real time. The dashboard will provide real-time log monitoring, alerts, and security analytics to help detect cyber threats.

Key Features & Functionalities

1. Real time logging monitoring

- System Logs : Track OS & application activities (login attempts, system failures).
- Network Logs : Monitor incoming & outgoing traffic, firewall activities.
- User Activity Logs : Record user authentication, failed login attempts.

2. Threat Alerts & Incident Detection

- Real-Time Security Alerts : Detect unauthorized access, malware, and brute-force attempts.
- Email & Push Notifications : Notify admin when a critical security event occurs.
- Risk Categorizations : High, Medium and Low severity classification.

3. Security Analytics & Visualization

- Live Dashboard with Graphics & Charts : Display attack trends, failed login attempts, malware detections.
- Heatmaps & Traffic Analysis : Monitor suspicious IPs, unusual login locations.
- Historical Data Analytics : Check past-security events for forensic investigations.

4. User Management & Role-Based Access

- Admin Panel : Manage user access, permissions, and security roles.
- Multi-Factor Authentication : Enhance Security with extra authentication layers

Tech Stack :

- ✓ **Backend** → Python + Flask (Lightweight, fast and easy to build REST APIs)
- ✓ **Frontend** → React.js (best for dynamic dashboards with real time updates)
→ Chart.js (for displaying security analytics (graph & charts))
- ✓ **Database** → MongoDB (flexible, handles unstructured security logs efficiently)

Additional tools :

- ✓ Celery + Redis → For processing security alerts in real time
- ✓ Docker → For easy deployment and scaling

Design : The UI will look something like this.



Cybersecurity Dashboard with Risk and Compliance

This graph/chart is linked to excel, and changes automatically based on data. Just left click on it and select "edit data".



Outcomes :

- ✓ Centralized security monitoring system for tracking cybersecurity threats.
- ✓ Automated alert system for real-time incident response.
- ✓ Data-driven security analytics to analyze trends and prevent future attacks.
- ✓ User-friendly dashboard for IT security teams.

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~Task 3 Week 5 (II)